Subtracting Across Zeros

No tens to break? Think of the number as tens and ones.

403 − 157: 3 − 7? Break a ten. But there are 0 tens!



4 hundreds, 3 ones

Break a hundred into 10 tens, then a ten into 10 ones.

The shortcut: 403 is 40 tens and 3 ones

403 = 40 tens + 3 ones

403 = 39 tens + 13 ones

Take 1 ten from the 40 tens. Now there are 39 tens and 13 ones. One break!

Now subtract: 157 is 15 tens and 7 ones

tens	39 − 15 = 24 tens
ones	13 − 7 = 6
total	240 + 6 = 246

Check by adding:

246 + 157 = 403

Your turn!

302 − 145 = 157

break	302 = 29 tens + 12 ones
tens	29 − 14 = 15
ones	12 − 5 = 7
total	150 + 7 = 157

For teachers

Goal: subtract when the top number has zeros (2.NBT.7). Say "403 is 40 tens and 3 ones," then break one ten: 39 tens and 13 ones. Watch for 0 − 7 = 7 (the 0 can't give). Check every answer by adding.

Subtracting Across Zeros: Practice

Think of the top number as tens and ones. Break once.

Part 1 Tens and ones Write it as tens and ones, then break one ten.

1) 403 = 40 tens + 3 ones
403 = 39 tens + 13 ones

2) 702 = 70 tens + 2 ones
702 = 69 tens + 12 ones

3) 305 = 30 tens + 5 ones
305 = 29 tens + 15 ones

4) 600 = 60 tens + 0 ones
600 = 59 tens + 10 ones

5) 801 = 80 tens + 1 ones
801 = 79 tens + 11 ones

6) 506 = 50 tens + 6 ones
506 = 49 tens + 16 ones

7) 900 = 90 tens + 0 ones
900 = 89 tens + 10 ones

8) 204 = 20 tens + 4 ones
204 = 19 tens + 14 ones

Part 2 Break, then subtract The break row is done for you.

9) 402 − 135 = 267

break	402 = 39 tens + 12 ones
tens	39 − 13 = 26
ones	12 − 5 = 7
total	260 + 7 = 267

10) 703 − 248 = 455

break	703 = 69 tens + 13 ones
tens	69 − 24 = 45
ones	13 − 8 = 5
total	450 + 5 = 455

Subtracting Across Zeros: Practice (continued)

Part 2 Break, then subtract (continued)

11) **500 − 164 = 336**

break: 500 = 49 tens + 10 ones

tens: 49 − 16 = 33

ones: 10 − 4 = 6

total: 330 + 6 = 336

12) **610 − 253 = 357**

break: 610 = 60 tens + 10 ones

tens: 60 − 25 = 35

ones: 10 − 3 = 7

total: 350 + 7 = 357

Part 3 Fill in every box

13) **804 − 376 = 428**

break: 804 = 79 tens + 14 ones

tens: 79 − 37 = 42

ones: 14 − 6 = 8

total: 420 + 8 = 428

14) **301 − 127 = 174**

break: 301 = 29 tens + 11 ones

tens: 29 − 12 = 17

ones: 11 − 7 = 4

total: 170 + 4 = 174

15) **900 − 465 = 435**

break: 900 = 89 tens + 10 ones

tens: 89 − 46 = 43

ones: 10 − 5 = 5

total: 430 + 5 = 435

16) **520 − 148 = 372**

break: 520 = 51 tens + 10 ones

tens: 51 − 14 = 37

ones: 10 − 8 = 2

total: 370 + 2 = 372

Subtracting Across Zeros: Practice (continued)

Part 4 Stack it Cross out the tens, write the new tens and ones on top, then subtract.

17)

hundreds	tens	ones
4	9	13
~~5~~	~~0~~	~~3~~
− 2	7	6
2	2	7

18)

hundreds	tens	ones
3	9	10
~~4~~	~~0~~	~~0~~
− 2	1	8
1	8	2

19)

hundreds	tens	ones
6	12	10
~~7~~	~~3~~	~~0~~
− 3	6	4
3	6	6

20)

hundreds	tens	ones
8	9	11
~~9~~	~~0~~	~~1~~
− 4	5	2
4	4	9

Part 5 Zero in the tens No help this time.

21)

3	9	13
~~4~~	~~0~~	~~3~~
− 1	5	7
2	4	6

22)

5	9	15
~~6~~	~~0~~	~~5~~
− 2	3	8
3	6	7

23)

7	9	12
~~8~~	~~0~~	~~2~~
− 4	6	5
3	3	7

24)

6	9	14
~~7~~	~~0~~	~~4~~
− 3	2	9
3	7	5

25)

4	9	11
~~5~~	~~0~~	~~1~~
− 2	8	6
2	1	5

26)

8	9	16
~~9~~	~~0~~	~~6~~
− 5	4	7
3	5	9

27)

2	9	18
~~3~~	~~0~~	~~8~~
− 1	5	9
1	4	9

28)

1	9	17
~~2~~	~~0~~	~~7~~
− 1	4	8
	5	9

Part 6 Zero in the ones No help this time.

29)

4	10	10
~~5~~	~~1~~	~~0~~
− 2	3	4
2	7	6

30)

6	11	10
~~7~~	~~2~~	~~0~~
− 3	8	5
3	3	5

31)

5	13	10
~~6~~	~~4~~	~~0~~
− 1	7	2
4	6	8

32)

7	12	10
~~8~~	~~3~~	~~0~~
− 4	5	6
3	7	4

Subtracting Across Zeros: Practice (continued)

Part 6 Zero in the ones (continued)

33)
$$\begin{array}{r} {\scriptstyle 3\ \ 14\ \ 10} \\ \not{4}\ \not{5}\ \not{0} \\ -\ 1\ 8\ 7 \\ \hline 2\ 6\ 3 \end{array}$$

34)
$$\begin{array}{r} {\scriptstyle 8\ \ 10\ \ 10} \\ \not{9}\ \not{1}\ \not{0} \\ -\ 5\ 3\ 8 \\ \hline 3\ 7\ 2 \end{array}$$

35)
$$\begin{array}{r} {\scriptstyle 2\ \ 16\ \ 10} \\ \not{3}\ \not{7}\ \not{0} \\ -\ 1\ 9\ 4 \\ \hline 1\ 7\ 6 \end{array}$$

36)
$$\begin{array}{r} {\scriptstyle 1\ \ 15\ \ 10} \\ \not{2}\ \not{6}\ \not{0} \\ -\ 1\ 7\ 5 \\ \hline \square\ 8\ 5 \end{array}$$

Part 7 Two zeros No help this time.

37)
$$\begin{array}{r} {\scriptstyle 2\ \ 9\ \ 10} \\ \not{3}\ \not{0}\ \not{0} \\ -\ 1\ 2\ 8 \\ \hline 1\ 7\ 2 \end{array}$$

38)
$$\begin{array}{r} {\scriptstyle 4\ \ 9\ \ 10} \\ \not{5}\ \not{0}\ \not{0} \\ -\ 2\ 4\ 6 \\ \hline 2\ 5\ 4 \end{array}$$

39)
$$\begin{array}{r} {\scriptstyle 6\ \ 9\ \ 10} \\ \not{7}\ \not{0}\ \not{0} \\ -\ 3\ 5\ 1 \\ \hline 3\ 4\ 9 \end{array}$$

40)
$$\begin{array}{r} {\scriptstyle 3\ \ 9\ \ 10} \\ \not{4}\ \not{0}\ \not{0} \\ -\ 1\ 6\ 3 \\ \hline 2\ 3\ 7 \end{array}$$

41)
$$\begin{array}{r} {\scriptstyle 7\ \ 9\ \ 10} \\ \not{8}\ \not{0}\ \not{0} \\ -\ 4\ 7\ 2 \\ \hline 3\ 2\ 8 \end{array}$$

42)
$$\begin{array}{r} {\scriptstyle 5\ \ 9\ \ 10} \\ \not{6}\ \not{0}\ \not{0} \\ -\ 2\ 8\ 9 \\ \hline 3\ 1\ 1 \end{array}$$

43)
$$\begin{array}{r} {\scriptstyle 8\ \ 9\ \ 10} \\ \not{9}\ \not{0}\ \not{0} \\ -\ 5\ 3\ 7 \\ \hline 3\ 6\ 3 \end{array}$$

44)
$$\begin{array}{r} {\scriptstyle 1\ \ 9\ \ 10} \\ \not{2}\ \not{0}\ \not{0} \\ -\ 1\ 4\ 6 \\ \hline \square\ 5\ 4 \end{array}$$

Part 8 Mixed Break only when you need to.

45)
$$\begin{array}{r} 6\ 0\ 5 \\ -\ 3\ 0\ 2 \\ \hline 3\ 0\ 3 \end{array}$$

46)
$$\begin{array}{r} {\scriptstyle 7\ \ 9\ \ 13} \\ \not{8}\ \not{0}\ \not{3} \\ -\ 2\ 6\ 7 \\ \hline 5\ 3\ 6 \end{array}$$

47)
$$\begin{array}{r} {\scriptstyle 3\ \ 10} \\ 5\ \not{4}\ \not{0} \\ -\ 3\ 1\ 5 \\ \hline 2\ 2\ 5 \end{array}$$

48)
$$\begin{array}{r} {\scriptstyle 6\ \ 9\ \ 10} \\ \not{7}\ \not{0}\ \not{0} \\ -\ 4\ 2\ 5 \\ \hline 2\ 7\ 5 \end{array}$$

Subtracting Across Zeros: Practice (continued)

Part 8 Mixed (continued)

49)

	3	9	16
	~~4~~	~~0~~	~~6~~
−	1	3	8
	2	6	8

50)

	8	16	
	~~9~~	~~6~~	0
−	5	8	0
	3	8	0

51)

	2	9	12
	~~3~~	~~0~~	~~2~~
−	1	8	5
	1	1	7

52)

		4	10
	8	~~5~~	~~0~~
−	6	2	4
	2	2	6

Part 9 Check by adding No help this time. Do you get the top number back?

53) 246 + 157 = 403

54) 177 + 128 = 305

55) 365 + 245 = 610

56) 226 + 274 = 500

57) 318 + 282 = 600

58) 459 + 141 = 600

Part 10 Remember? Problems from earlier packets.

59) 65¢ + 25¢ = 90¢

60) half past 7 = 7:30

61) 352 + 216 = 568

62) 64 + 36 = 100

63) half of 86 = 43

64) 2 halves = 1 whole

Part 11 Word problems Break once. Check by adding.

65) A school has 400 pencils. The kids use 156 of them. How many pencils are left?

Number sentence 400 − 156 = 244

Answer 244 pencils

Draw or show your work

Subtracting Across Zeros: Practice (continued)

Part 11 Word problems (continued)

66) A book has 305 pages. Mia has read 128 pages. How many pages are left to read?

Number sentence **305 − 128 = 177**

Answer **177** pages

Draw or show your work

67) There were 610 people at a game. Then 245 people went home. How many people stayed?

Number sentence **610 − 245 = 365**

Answer **365** people

Draw or show your work

68) A jar holds 500 beads. 274 beads are red and the rest are blue. How many beads are blue?

Number sentence **500 − 274 = 226**

Answer **226** beads

Draw or show your work

69) Challenge: A game needs 1000 points. Kim has 455. How many more points does she need?

Number sentence **455 + 545 = 1000**

Answer **545** points

Draw or show your work

Check-In: Subtracting Across Zeros

Show what you know. No help this time.

Mastery: 8 of 9 (80%)

Part 1 Tens and ones

1) 903 = 90 tens + 3 ones

903 = 89 tens + 13 ones

2) 409 = 40 tens + 9 ones

409 = 39 tens + 19 ones

Part 2 Subtract Break once.

3)

	5	10	10
	~~9~~	~~1~~	~~0~~
−	4	7	3
	1	3	7

4)

	7	9	12
	~~8~~	~~0~~	~~2~~
−	2	1	3
	5	8	9

5)

	8	16	10
	~~9~~	~~7~~	~~0~~
−	6	8	1
	2	8	9

6)

	7	12	10
	~~8~~	~~3~~	~~0~~
−	7	3	3
		9	7

Part 3 Check by adding

7) 268 + 632 = 900

8) 85 + 615 = 700

9) 464 + 356 = 820